

Technical Requirements Document

Seafood Marketplace – Single Page E-Commerce Website

Version: 1.0

Date: March 12, 2026

Audience: Full Stack Engineers and Technical Leads

1. System Overview

The Seafood Marketplace website will function as a Single Page Application (SPA) allowing customers to browse seafood products, place pickup orders, and request catering services. The system focuses on fast navigation, responsive design, and simplified checkout.

2. Navigation Requirements

Top Navigation Links:

- Home
- About Us
- Products
- Catering
- Testimonials
- Login (Icon) – aligned right
- Cart (Icon) – aligned right

Important Note: Remove the 'Book Catering' button and replace it with a 'Catering' navigation link.

3. Catering Form Requirements

Fields required for the catering form:

- Name
- Email Address
- Phone Number
- Event Type
- Message

4. Authentication Requirements

Users must be able to login or register as a customer.

After successful authentication users will be redirected to a customer dashboard where they can view:

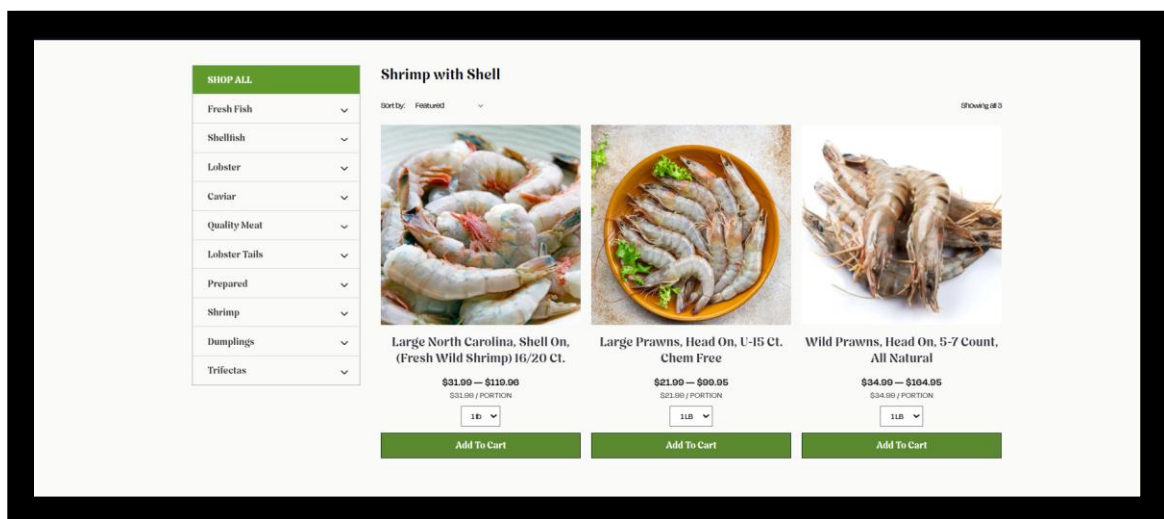
- Order History
- Status of Current Orders

5. Products and Product Details

When a user clicks a seafood product they are directed to a Product Details page containing:

- Product Image
- Product Description
- Product Price
- Weight Selection Dropdown (lbs)
- Add to Cart Button

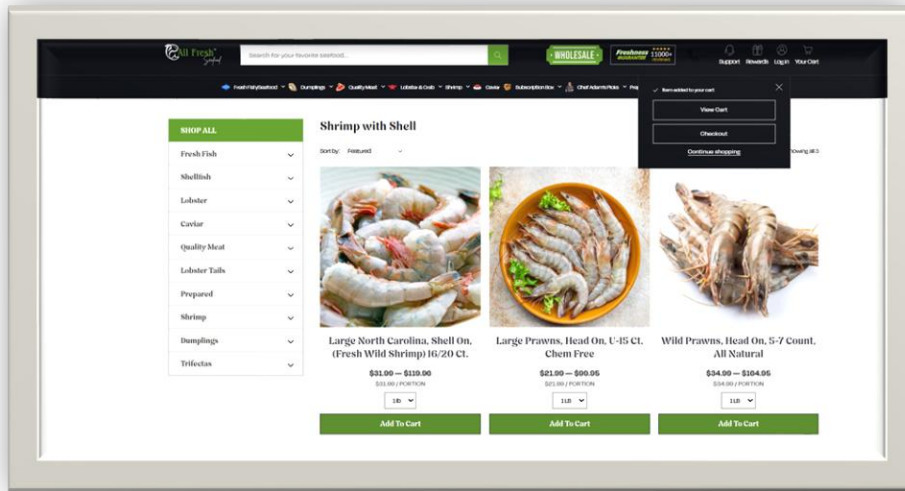
The product details page must follow a horizontal three-column layout:



6. Add to Cart Workflow

When a user clicks 'Add to Cart', a modal or screen should display the following options:

- View Cart
- Check Out
- Continue Shopping



7. Checkout Requirements

The checkout page must collect customer and order information including:

- Customer Name
- Email Address
- Phone Number
- Pickup Location
- Products Purchased
- Payment Details

Pickup Location Options: East, South, West

Payment Methods: Credit Card or Afterpay

The checkout screen must include a final 'Pay Now' button to complete the transaction.

City State ZIP code

Phone

☐ Text me with news and offers

Shipping method

Enter your shipping address to view available shipping methods.

Payment
All transactions are secure and encrypted.

☒ Credit card VISA +5

Card number

Expiration date (MM / YY) Security code

Name on card

☒ Use shipping address as billing address

☐ Shop Pay • Pay in full or in installments shop

☐ PayPal PayPal

Product List:

- Large North Carolina, Shell On, (Fresh Wild Shrimp) 16/20 ct. 1 lb \$31.99/lb \$31.99

[Don't miss out! Log in to earn and redeem rewards](#)

Discount Code, Gift Card or Store Credit Apply

Subtotal \$31.99
Shipping Enter shipping address

Total USD **\$31.99**

11,000+ Happy Customers 5 stars Sustainably Sourced Seafood leaf icon We Support U.S. Veterans and First Responders US flag icon

18. Pre-Order Functionality

18.1 Overview

The system must support **Pre-Orders** for seafood products that may not be immediately available or are reserved for a future pickup date.

Pre-orders allow customers to **reserve seafood inventory in advance** by placing an order and paying a **required deposit**. This ensures product availability and reduces inventory risk for the business.

Examples of products eligible for pre-order:

- Snow Crab
- King Crab

- Lobster
- Crawfish (seasonal)
- Bulk seafood catering trays

Pre-orders are typically used for:

- **Large quantity purchases**
- **Seasonal seafood**
- **Holiday or weekend demand**
- **Catering events**

18.2 Pre-Order Indicators in Product Catalog

Products that allow pre-orders must display a **visual indicator** on the product card.

Example badge:

PRE-ORDER AVAILABLE

Product cards should also display:

Available for Pickup: [Pickup Date]

Deposit Required

Example:

Snow Crab

\$28 / lb

Available for Pickup: April 5

Deposit Required

18.3 Pre-Order Product Page Behavior

When a product supports pre-orders, the **Product Details Page** must include additional fields.

Additional Fields

Field	Type	Required
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	Field	Type	Required
Date	Pickup	Date	Y
	Picker	es	
(lbs)	Weight	Drop	Y
	down	es	
y	Quantit	Num	Y
	ies	es	
Amount	Deposit	Cal	Y
	culated	es	

Example UI:

Weight: [Dropdown]
Quantity: [1]
Pickup Date: [Calendar]

Deposit Due Today: \$50
Remaining Balance: \$150

18.4 Pre-Order Deposit Business Rules

Pre-orders require a **deposit payment** at checkout.

Deposit Rules

Rule	Description
Deposit Required	Yes
Deposit Percentage	25% – 50% of total order
Remaining Balance	Paid at pickup
Payment Methods	Credit Card / Afterpay

Example:

Order Total: \$200

Deposit (25%): \$50

Remaining Balance: \$150

Deposit ensures:

- Reservation of seafood inventory
 - Reduced no-shows
 - Guaranteed pickup allocation
-

18.5 Deposit Calculation Logic

Example calculation:

$\text{deposit_amount} = \text{order_total} * \text{deposit_percentage}$

$\text{remaining_balance} = \text{order_total} - \text{deposit_amount}$

Example:

$\text{order_total} = \$240$

$\text{deposit_percentage} = 0.25$

$\text{deposit_amount} = \60

$\text{remaining_balance} = \180

Only the **deposit amount** is charged during checkout.

18.6 Pre-Order Checkout Flow

The checkout process for pre-orders differs slightly from regular orders.

Pre-Order Workflow

1. Customer selects product
2. Customer selects weight and quantity
3. Customer selects **pickup date**
4. System calculates **deposit**
5. Customer proceeds to checkout

6. Customer pays **deposit**
7. Order is recorded as **Pre-Order**

Order status:

Pre-Order Reserved

18.7 Pre-Order Confirmation

After payment the system must show confirmation.

Example:

Thank you for your Pre-Order!

Order Number: #10823

Product: Snow Crab

Pickup Date: April 5

Deposit Paid: \$60

Remaining Balance Due at Pickup: \$180

Customer also receives **email confirmation**.

18.8 Information Required for Pre-Order Processing

The system must capture the following information.

Customer Information

Field	R equired
Full Name	Y
Email Address	Y
Phone Number	Y

Order Information

	Field	R equired
Name	Product	Y
	es	
Selected	Weight	Y
	es	
y	Quantit	Y
	es	
Date	Pickup	Y
	es	
Location	Pickup	Y
	es	

Pickup Locations:

East

South

West

Payment Information		
	Field	R equired
Method	Payment	Y
	es	
	Deposit Amount	Y
	es	
Transaction ID	Payment	Y
	es	

Payment options:

Credit Card

Afterpay

18.9 Order Status for Pre-Orders

Pre-orders must support the following statuses.

Status	Description
Pre-Order Reserved	Deposit paid
Processing	Order being prepared
Ready for Pickup	Available for customer
Completed	Order picked up
Cancelled	Order cancelled

18.10 Database Changes for Pre-Orders

The **orders table** must include additional fields.

Example:

orders

id
customer_id
order_type (regular | preorder)
order_total
deposit_amount
remaining_balance
pickup_date
pickup_location
status
created_at

Order Items Table

order_items

id
order_id

product_id
weight
quantity
price

18.11 Admin Requirements

Admins should be able to:

- View all pre-orders
- Filter by pickup date
- Track deposits
- Mark orders ready for pickup

Recommended dashboard widget:

Pre-Orders by Pickup Date

18.12 Edge Cases

Engineers must account for:

Inventory Limits

System should prevent orders exceeding available inventory.

Missed Pickup

If a customer fails to pickup:

Deposit is non-refundable

Business rule configurable.

Cancellation Window

Allow cancellation within:

24 hours of placing the pre-order

18.13 Benefits of Pre-Order System

Implementing pre-orders enables the business to:

- Forecast seafood demand
- Reduce waste
- Secure inventory purchases from suppliers
- Improve weekend and holiday sales
- Guarantee product availability for customers